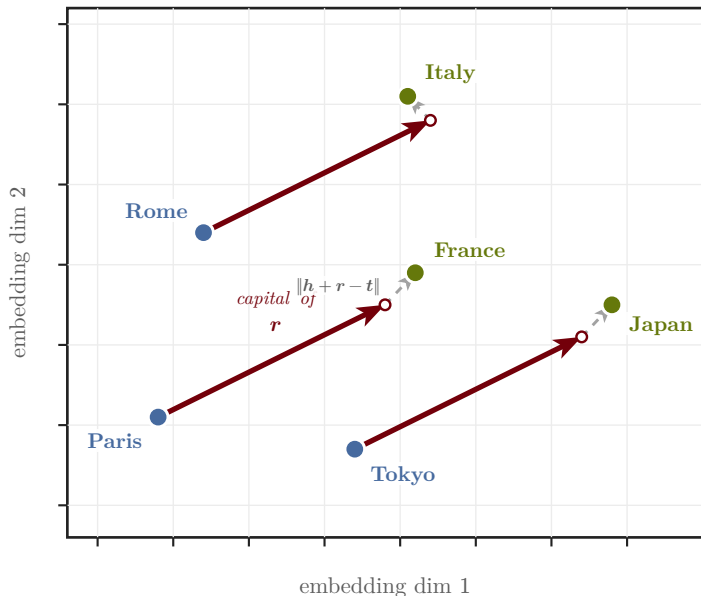


Knowledge-graph embedding — TransE

$$h + r \approx t$$

one relation = one translation vector, shared across all its triples



Triples (h, r, t)

head h	relation r	tail t
Paris	<i>capital of</i>	France
Rome	<i>capital of</i>	Italy
Tokyo	<i>capital of</i>	Japan

Scoring a triple

$$f(h, r, t) = -\|h + r - t\|_{1/2}$$

high score (~ 0): true triple

low score: corrupted triple (h, r, t')

Margin loss

$$\mathcal{L} = \sum [\gamma + d(h\{+\}r, t) - d(h\{+\}r, t')]_+$$

push true tail closer than a corrupted tail t' by margin γ

 relation translation r

 residual $h\{+\}r\{-}t$

 head entity (capital)

 tail entity (country)